

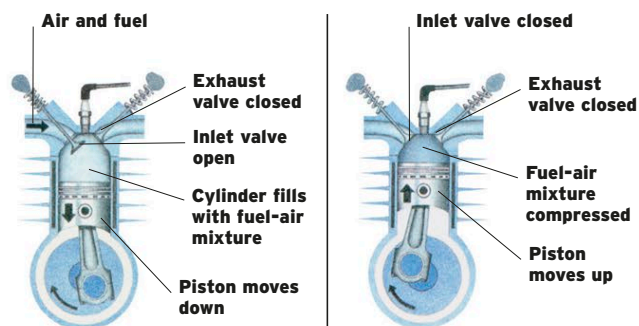
Piston Engines

It was the appearance of the gasoline engine late in the 19th century that made powered flight possible at last: pioneers had experimented with steam engines, but even the lightest of these proved too weighty. Early car engines produced sufficient power but were too heavy. So while they could operate on the same principles as car engines, piston aero engines had to be as light as possible. They also had to be reliable, and these requirements favored special light alloy

construction and dual magneto ignition. Because it kept engines simple and light, air-cooling became the choice for light aircraft and commercial transport—but for half a century or more battle raged over whether or not liquid-cooled engines offered high-speed aircraft more power for less drag. While the appearance of the jet engine more or less ended that debate, the liquid-cooled aero engine has recently found favor in ultralights and Light Sport Aircraft.

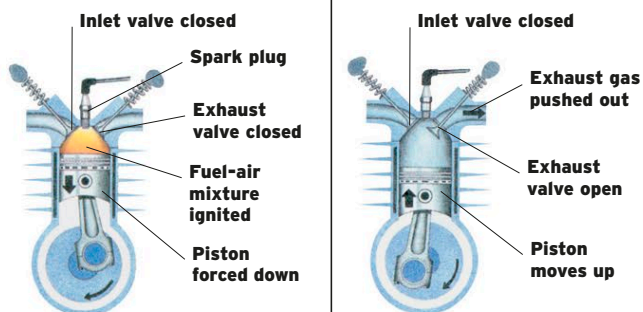
FOUR-STROKE CYCLE

Piston engines produce their power through the pressure of burning gases on the piston crowns acting on a rotating crankshaft. In the classic four-stroke Otto cycle (named after its inventor and illustrated below), the spark plug is fired every other rotation of the crankshaft. Virtually all piston aero engines are four-strokes, and work just like car engines, although they are generally of bigger cylinder capacity to compensate for turning a propeller no more than 3,000 rpm. From the 1930s military and commercial aircraft engines were fitted with superchargers, cramming in more mixture to sustain power in the thin air at high altitude.



1 Induction The inlet valve opens and the piston moves down, drawing the fuel-air mixture into the cylinder through the engine's inlet and fueling system.

2 Compression The piston moves back up the cylinder. This increases the pressure inside the cylinder, heating the fuel-air mixture.



3 Expansion When the piston is near the top of its stroke, a spark plug fires. The burning gas expands, forcing the piston down the cylinder again.

4 Exhaust As the piston reaches bottom dead center, the exhaust valve opens. As the piston rises again, it forces waste gases out of the exhaust.

ROLLS-ROYCE MERLIN

For all its apparent complexity, the Rolls-Royce Merlin has pistons and valves just like those of a car engine—only more of them. The amount of power produced by an engine is proportional to how fast it runs and how much air can be drawn into it. The Merlin was made to run fast by gearing down the propeller: its supercharger pushed in extra air, boosting power at altitude.

Inlet manifold sends charge to all 12 cylinders

Supercharger forces more mixture through the engine, maintaining power at altitude

